4.5, 7, 11, 12.8

NED University of Engineering and Technology
Department of Computer and Information Systems Engineering
CS-323 Artificial Intelligence
Midterm Examination - Fall Semester 2025
TE (CIS) Batch 2023

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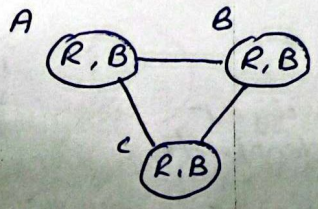
Marks Obtained: 34.5
Max. Marks: 40

Question 1 [CLO-1, C3]

1.a AC-3 can always find solution for CSPs without using backtracking search. [2]

AC3 cannot always find solution even if it is consistent.
However, it doesnot require backtracking and can yield the solution if exists.

Eg:



consistent yet no solution.

1.b Local search and A* search both are complete and optimal. [2]

Local search is not always complete and optimal as it may skip the critical point due to larger steps and can take huge amount of time due to small steps

2

A* search is complete if ε (minimum step) > 0 and b (branching factor) is finite.
And is optimal when the heuristics are consistent and admissible.

1.c Goal-based agents are more flexible in their decision-making compared to model-based agents. [2]

model based agents have memory and are biased?
whereas goal based agents looks for the path leading to goal without thinking about past experience.
Hence, Goal based agents are more flexible.

1

Question 2 [CLO-1, C3]

Consider a software agent trained to identify the following nine objects:
Plane, Car, Bird, Cat, Deer, Dog, Frog, Horse, Ship and Truck
A user can give as input an image of any of these nine objects and the agent tries to correctly identify it.

2.a Write PEAS description for this software agents. [2]

2.b Identify the following properties for this software agent. Also justify your selection. [3]

- Partially observable or fully observable
- Single-agent or multi-agent
- Deterministic or stochastic
- Sequential or episodic
- Dynamic or Static
- Discrete or continuous

(2.a)

Performance measure:
→ correctly identify object in the image.
→ minimize the non-identification

Environment:
→ Picture, data of objects (plane, car, bird, ... etc) ~~that~~ the nine objects.

Image ← Sensors:
→ ~~Camera~~
→ pixels mapping
→ features of object mapping

Actuators:
→ display the name of object
→ display that not identified.

4

(2.b) ~~Fully Observable~~ Partially ~~Fully~~ Observable:
As the input will always be any of the nine objects.

2b) Partially Observable:
It maybe given the image that it was not trained with.
(eg: Dog with fur)

Single Agent: The only agent is software agent.

~~Deterministic: Either it will identify or not. The pictures will be of any of 9 objects.~~ Stochastic: It can give answer in percentage. eg: 10% like bird, 80% like plane

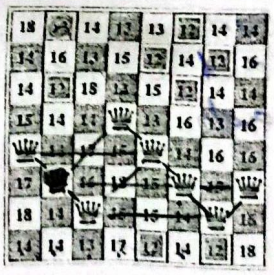
Episodic: Independent of previous identification.

Static: One picture will be shown at a time.

~~Discrete: The output and input both are discrete.~~ Continuous: The output can be of various range.

Question 3 [CLO-1, C3]

The given figure shows an 8-queens state with heuristic cost estimate h=17. The board also shows the heuristic value h for each possible successor obtained by moving a queen within its column. The heuristic used is the number of attacking pairs of queens, formulated as a minimization problem. Determine the successor h value which will be chosen if we apply:



3.a deterministic hill climbing. [3]

3.b simulated annealing (assume a nonzero temperature T). [3]

(3.a)

~~Q2 also had 4 attackers but we choose Q2~~

h values:
$Q_1 = 3$
$Q_2 = 4$
$Q_3 = 2$
$Q_4 = 3$
$Q_5 = 2$
$Q_6 = 2$
$Q_7 = 1$
$Q_8 = 0$
$h = 17$

Q_2 has highest no. of attacking queens.
if we move Q_2 to row 1 new h value
$Q_1 = 2$ $Q_2 = 0$ $Q_3 = 2$
$Q_4 = 3$ $Q_5 = 2$ $Q_6 = 2$
$Q_7 = 1$ $Q_8 = 0$

Summing them all:

$\boxed{h_{new} = 12}$

(3.b)

~~let T = 0.2 (random~~

The random choosen value:
Q_1 = (row) 3

$\boxed{\text{Hence, the succe value of } h = 14}$

3

2

Question 4 [CLO-2, C3]

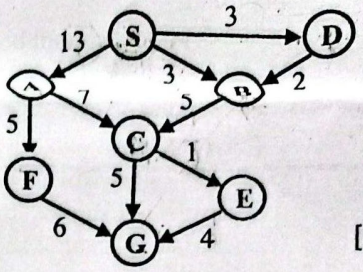
S→G UCS

For the given graph:

4.a Work out the order in which states are expanded and explored for the path from node **S** to node **G** using Uniform Cost Search. Show all steps of the algorithm, including the frontier (priority queue) updates, and maintain an expanded list. Use the following tie-breaking rule: [5]

- If two nodes have the same path cost and are already in the queue, expand in First-Come First-Served (FCFS) order.
- If two nodes have the same path cost and are generated in the same step, order them alphabetically (lower alphabet gets priority).

4.b Can you use early goal testing in place of late goal testing? Briefly justify your answer. [2]



Q4(a)

F	Expanded
~~(0,S)~~	S
(13,SA)(3,SD)~~(3,SB)~~	SB
(13,SA)(8,SBC)~~(3,SD)~~	SBD # B expanded SBD not added
(13,SA)~~(8,SBC)~~	SBCD
(13,SBCG)~~(13,SA)~~(9,~~SBCE~~)	SBCDE
(13,SBCEG)(13,SBCG)~~(13,SA)~~	SBCDEA
(18,SAF)(13,SBCEG)~~(13,SBCG)~~	SBCEA

solution: SBCG
Path Cost: 13

5

Q4(b) No, early goal testing cannot be use as it doesnot guarantee lowest cost path. Your algo may return a higher cost path when there exist a lower cost path.

2

Question 5 [CLO-2, C3]

Recall the implementation of Gradient Descent algorithm discussed in class. Assume the following values for the arguments passed to the gradient descent algorithm. [4]

initial_value_θ_1 = 0.5, initial_value_θ_2 = 0.25, learning_rate = 0.02

Find value for θ_1 and θ_2 after first iteration if the cost function $J(\theta_1, \theta_2) = 4\theta_1^2 + \theta_2$. Show all your calculations.

$$\frac{\partial J(\theta_1,\theta_2)}{\partial \theta_1} = 8\theta_1$$

$$\frac{\partial J(\theta_1,\theta_2)}{\partial \theta_2} = 1$$

Putting respective values:

$$\frac{\partial J(\theta_1,\theta_2)}{\partial \theta_1} = 8(0.5) = 4$$

$$\frac{\partial J(\theta_1,\theta_2)}{\partial \theta_2} = 1$$

After first iteration:

$$\theta_1 = \theta_1 - \text{learning rate} \times \frac{\partial J(\theta_1,\theta_2)}{\partial \theta_1}$$

$$= 0.5 - 0.02(4)$$

$$\boxed{\theta_1 = 0.42}$$

4

$$\theta_2 = \theta_2 - \text{learning rate} \times \frac{\partial J(\theta_1,\theta_2)}{\partial \theta_2}$$

$$= 0.25 - (0.02 \times 1)$$

$$\boxed{\theta_2 = 0.23}$$

3

CamScanner

Question 6 [CLO-2, C3]

Six projects **G, H, I, J, K, L** must be supervised by one of three supervisors **P, Q, R**.

Constraints:

1. A project cannot be supervised by more than one supervisor. However, a supervisor can supervise more than one projects.
2. Project **J** must be supervised by **Q**.
3. Project **K** cannot share a supervisor with **J** or with **L**.
4. Anybody supervising **I or J cannot** supervise **L**.
5. The supervisor of **I** cannot supervise **J or H**.
6. Project **H** cannot share a supervisor with **G**.
7. Project **G** cannot share a supervisor with **J**.

6.a	Formulate this as a constraint satisfaction problem (CSP) by identifying variables and their domains.	[3]
6.b	Identify the constraints as unary, binary and higher order constraints.	[2]
6.c	Draw constraint graph and solve the formulated CSP by backtracking with forward checking.	[5]
6.d	Is your solution unique? If not, then give at least one other solution. No need to show solution again.	[2]

You may use at most 1 additional sheet (2-sided) to answer this question.

6.a:

Variables = G, H, I, J, K, L

Domains =

$D_G = \{P, Q, R\}$ $D_H = \{P, Q, R\}$

$D_I = \{P, Q, R\}$ $D_J = \{Q\}$

$D_K = \{P, Q, R\}$ $D_L = \{P, Q, R\}$

3

6.b Constraints:

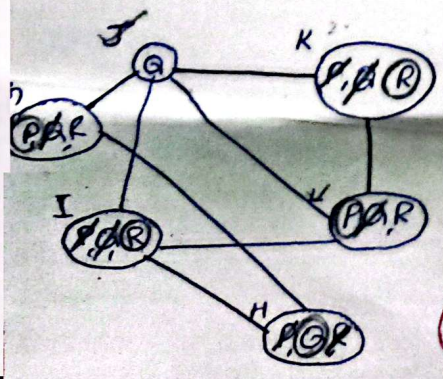
Unary = $J = Q$

Binary = $K \neq J$, $K \neq L$, $L \neq I$, $L \neq J$
$I \neq J$, $I \neq H$, $H \neq G$, $G \neq J$

Higher Order = $K \neq J, L$
$L \neq I, J$
$I \neq J, H$

2

6.c: Constraint Graph:



$J = Q$
$K = R$
$L = P$
$H = Q$
$I = R$
$G = P$

5

6.d

No, my solution is not unique

Another solution:

$J = Q$ $K = R$ $L = P$ $H = P$ $G = R$

$I = R$

0.5 + 1.5

(G)

(P&R)